TICCO ROSS

EX-800

8/16-voice programmable synthesizer · AU / VST3

User manual · version 0.6.0



A reconstruction of a classic 1980s digital-analog hybrid: eight (or sixteen) voices of stacked square-wave footages, three six-stage envelopes, one shared resonant filter, and stereo chorus, phaser and ensemble.

1. Getting started

Installing

- Run the **EX-800 .pkg** installer. It installs the Audio Unit to */Library/Audio/Plug-Ins/Components* and the VST3 to */Library/Audio/Plug-Ins/VST3*.
- The installer is not yet signed by Apple. If macOS blocks it, right-click the .pkg and choose **Open**, or allow it under **System Settings › Privacy & Security**.
- In Ableton Live, EX-800 appears under **Plug-ins › Audio Units › Ticco Ross** (or VST3). Rescan plug-ins if it does not show up.
- EX-800 runs at any sample rate. Below 88 kHz it runs its engine at twice the host rate internally, so it sounds the same at 44.1, 48 and 96 kHz.

First sounds

EX-800 opens on factory program **11 Strings**. Step through the 64 factory programs with the **< PROG** and **PROG >** buttons under the display, from your host's preset menu, or by typing a program number on the keypad (section 3).

2. The panel



Section	Controls
Joystick / Tune	**Tune** slider (±100 cents, double-click to reset). **Joystick**: left/right bends pitch, up adds vibrato (MG to DCO), down adds filter wobble (MG to VCF). It springs back when released. **BEND RANGE** < / > under it sets how far it bends (±1 to ±12 semitones).
Volume	Master output level.
Key assign	**POLY**: normal playing. **CHORD**: chord memory, each key plays a stored chord (section 4). **HOLD**: notes keep sounding after you let go (the sustain pedal does the same). **CHORD MEMORY** stores the keys you are holding as the chord.
Programmer	LED display (program, parameter, value), keypad, PROG/PARA key, **DOWN** / **UP**. **ABOUT** (or a click on the EX-800 name) shows the version and contact details.

Section	Controls
Performance / Effects	**VOICES** switch (8 voices WHOLE / 4 voices DOUBLE, same as P18), **16 VOICES** (doubles the polyphony: 16 WHOLE / 8 DOUBLE), **EFFECT** on/off (P48), effect **TYPE** (Chorus, Phaser or Ensemble) and the **SPEED** slider (chorus or phaser rate, or ensemble intensity).
Presets	**BANK** (click to choose a bank), the preset name (click for the bank's list), **<** / **>** to step through it, **SAVE**, **NEW BANK**, **DELETE** and the favourite **star** (section 4).
Sequencer	**SEQUENCER** swaps the parameter chart for the step sequencer screen (section 5); press it again for the chart. **PLAY** starts the sequencer and reads **STOP** while it runs.
Parameter chart	Every sound parameter, laid out like the printed chart on the original. Each cell shows its name, number, range and current value.

3. Editing sounds

The original instrument has no knobs: you pick a parameter by number and change its value. EX-800 keeps that workflow and adds direct mouse editing on the chart. All methods change the same parameters.

On the chart

- **Click** a cell to select it. Its number and value appear on the LED display.
- **Drag** a cell up or down to change its value, or use the **scroll wheel** over it.
- Press **UP** / **DOWN** to step the selected parameter one value at a time.

With the keypad

- The keypad has the digits 1-8, as on the original, plus the **PROG / PARA** key.
- In **PARA** mode, type two digits to select a parameter: **4** then **1** selects P41 Cutoff.
- Press **PROG/PARA** to switch to **PROG** mode. Two digits now load a factory program: **2** then **4** loads 24 “1999”.
- After the first digit the display shows it with a dash, waiting for the second.

Saving your sounds

Your edits are stored with your Live set like any plug-in. To keep a sound for other projects, save it as a EX-800 preset (section 4, Presets and banks).

4. Playing

Joystick and controllers

- **Joystick X** bends pitch by up to ±**Bend Range** semitones: 1-12, default 2. Set it with the **BEND RANGE** < / > buttons under the joystick (it is also a host parameter, and is saved in presets).
- **Joystick up** adds MG vibrato to the DCOs; **down** adds MG movement to the filter. Both add to the program's own MG depths (P83/P84) and follow the MG speed (P81).
- Your keyboard's **pitch wheel** adds to joystick X, and its **mod wheel** (CC1) works like pushing the joystick up.
- Joystick X and Y are host parameters, so you can automate them.

Presets and banks

Presets are organised in banks, like the programs of the original. The preset section is under the effect speed slider:

- **Factory** is the bank of the 64 original programs, 11 to 88. It is built in and cannot be changed.
- **User** is the first bank for your own sounds. Make more with **NEW BANK**, one per project or style.
- **Favourites** lists every preset you have starred, from any bank.
- Click **BANK** to choose a bank (its first preset loads). Click the preset name for the full list of the bank, or step through it with **<** and **>** (the < PROG and PROG > buttons do the same).
- Presets in every bank are numbered like the hardware: 11-18, 21-28 up to 88. The PROGRAM NO. display shows the number, and in PROG mode the keypad loads a preset by number from the bank shown.

- **SAVE** stores the current sound: give it a name and choose the bank. Saving under an existing name asks before replacing it. A preset stores all the sound parameters, the effect type and speed, the bend range, the key assign mode, the chord memory and the sequencer's eight patterns (not volume, tune or the joystick). Loading a factory program keeps the patterns you have.
- **DELETE** removes the current preset (user banks only). The bank menu can delete a whole bank, open a preset file from anywhere, or show the preset folder.
- The **star** adds or removes the current preset from Favourites.

Preset files (.p800preset) are kept in one folder per bank. macOS: *~/Library/Audio/Presets/Ticco Ross/EX-800*. Windows: *Documents\Ticco Ross\EX-800\Presets*. Linux: *~/.local/share/Ticco Ross/EX-800/Presets*. Presets saved by P-800 (the earlier name) are moved there automatically. Copy a bank folder there to share it.

Chord memory

Like the original's chord memory, CHORD mode plays a whole stored chord from a single key. It works the same way as in Polytwelve:

- Hold the chord you want (two or more keys) and press **CHORD**. The keys are stored, lowest note first.
- Or, already in CHORD mode, hold the keys and press **CHORD MEMORY**. With HOLD on you can play the notes one at a time first, then press CHORD MEMORY.
- Now each key plays the stored chord transposed to it. The most recent key wins (last-note priority); release it and the chord returns to the key still held.
- The stored chord and its name appear in the top strip, and it is saved with your session. Pressing CHORD with fewer than two keys held keeps the previous chord.
- A chord can have as many notes as there are voices: 8 in WHOLE mode, 4 in DOUBLE (16 and 8 with 16 VOICES on); extra notes are dropped from the top.
- Press **POLY** to go back to normal playing (POLY also switches HOLD off).

Effects

EX-800 has the same three stereo effects as Polytwelve. P48 or the EFFECT switch turns the effect on and off; **TYPE** chooses it:

- **Chorus** (default): a BBD-style stereo chorus. SPEED sets its rate, about 0.1 to 4 Hz.
- **Phaser**: a six-stage stereo phaser with feedback. SPEED sets its rate, about 0.05 to 6 Hz.
- **Ensemble**: a lush three-voice string ensemble. The slider sets its intensity.

Voices and the shared filter

WHOLE mode gives 8 voices with DCO 1 only; DOUBLE mode gives 4 voices, each with DCO 1 and DCO 2 for fatter, detuned or interval sounds. **16 VOICES** doubles both: 16 voices in WHOLE mode, 8 in DOUBLE (the VOICES switch labels change to show it). Changing it silences the notes sounding. All voices go through **one shared filter**, as on the original, so a new note can retrigger the filter envelope for chords already sounding (see P46 Trigger). Keyboard tracking follows the highest note held.

5. Step sequencer

A polyphonic step sequencer of 16 or 32 steps, with eight patterns (A to H) and parameter locks. Press **SEQUENCER** to show it in place of the parameter chart, and **PLAY** to run it. It keeps running when you go back to the chart.



Top bar

- **LENGTH** 16 or 32 steps. Going from 16 to 32 with an empty second half copies the first half into it.
- **PATTERN** A to H. Patterns that hold something read brighter. While the sequencer is stopped a new pattern is chosen at once. While it plays, the change is **queued**: the chosen pattern blinks and the playing one (marked with an amber bar) finishes its last step first; the new pattern then starts from step 1.
- **RATE** (click): step length, 1/8, 1/16, 1/32, 1/8T or 1/16T. **SWING** 50% (straight) to 75% delays every second step. **GATE** is how much of its last step a note sounds. Drag up/down or scroll to change them.
- **SYNC**: **HOST** follows your DAW's tempo and transport (the sequencer plays while the DAW plays, if PLAY is on); **INT** runs on its own clock at **BPM**, from step 1 when you press PLAY.
- **OCT** scrolls the grid by an octave. **REC** records from your keyboard (below). **COPY** copies the pattern and becomes **PASTE**: choose another pattern and press it. **CLEAR** empties the pattern.
- **TIE** and **REST** are for step recording (below).

Notes

- The grid shows two octaves; each column is a step. **Click** an empty cell to add a note, **drag** to the right to make it longer, and click a note to remove it.
- Steps are polyphonic: up to 8 notes each (in DOUBLE mode the synth has 4 voices, or 8 with 16 VOICES on, so bigger chords lose notes).
- Clicking a step (or its number) selects it, outlined in the numbers row.

- **Live recording**: with REC on while the sequencer plays, the notes you play go on the nearest step and last as long as you hold them, so sustained notes are recorded as long notes (with HOLD or the sustain pedal they last until you release it).
- **Step recording**: with REC on and the sequencer stopped, play a note or chord: it goes on the selected step, and when you let go of all keys the selection moves to the next step. Press **TIE** to stretch the notes you just entered over the next step (press it again for more steps), or **REST** to leave a step empty.
- The sequencer records what the key assign plays: in CHORD mode one key records the whole stored chord, and with HOLD on you can build a chord note by note.

Lanes: probability, gate and parameter locks

- **PROB**: the chance (0-100%) that a step plays. Drag a bar; right-click or double-click resets it to 100%.
- **GATE**: how long the step's notes sound, as a percentage of the global GATE.
- **P-LOCK**: choose a parameter at the top of the lane, then drag a bar to lock that parameter to a value on that step. Any chart parameter can be locked, and so can the other assignable controls (in the *Effect / Performance* part of the menu): Effect Speed, Effect Type, Volume, Tune and the joystick (X bends the pitch, Y adds vibrato or filter movement). The menu marks parameters that have locks. Steps without a lock show a dash and use the panel value. Right-click or double-click removes a lock.
- **Resizing the lane**: drag the handle between the note grid and the lane up to make the lane taller (for finer values), or down to give the grid its room back. With a tall lane the grid shows one octave; use OCT to scroll it. The size is remembered with your Live set.
- A step can lock several parameters: choose another parameter and draw its values. A red dot on a step number means the step has locks. Locks hold for the length of their step.

The Poly-800 has no velocity, so notes always play at the same level; use the filter and envelope parameters (or locks on them) for accents.

6. Parameter reference

DCO 1

No.	Name	Range	What it does
11	Octave	1-3	Pitch register of DCO 1: 1 low (16'), 2 middle, 3 high.
12	Waveform	1-2	1 square. 2 the stepped saw-like mix (footages weighted 1, 1/2, 1/4, 1/8).
13-16	16' / 8' / 4' / 2'	0-1	Switch each square-wave footage on or off. Combine them to build the tone.
17	Level	0-31	DCO 1 volume.
18	Mode	1-2	1 WHOLE: 8 voices (16 with 16 VOICES on), DCO 1 only. 2 DOUBLE: 4 voices (8), DCO 1 + DCO 2.

DCO 2 (heard in DOUBLE mode)

No.	Name	Range	What it does
21	Octave	1-3	Pitch register of DCO 2.
22	Waveform	1-2	As P12.
23-26	16' / 8' / 4' / 2'	0-1	DCO 2 footages.
27	Level	0-31	DCO 2 volume.
31	Interval	0-12	DCO 2 transposition in semitones above DCO 1.
32	Detune	0-3	Detunes DCO 2 downward, up to about -20 cents.

Noise

No.	Name	Range	What it does
33	Noise level	0-15	White noise into the filter, shaped by DEG 3. At 15 it is about as loud as one full voice.

VCF (one filter shared by all voices)

No.	Name	Range	What it does
41	Cutoff	0-99	Filter cutoff frequency.
42	Resonance	0-15	Emphasis at the cutoff. Very high settings ring and whistle; at 15 the filter self-oscillates while notes sound.
43	Keyboard track	0-2	0 off, 1 half, 2 full: the cutoff rises with the highest note played (full: one octave per octave, from middle C).
44	EG polarity	1-2	1 inverted, 2 normal direction of the DEG 3 sweep.
45	EG intensity	0-15	How far DEG 3 sweeps the cutoff.
46	Trigger	1-2	1 single: DEG 3 restarts only when all keys are released. 2 multi: every note restarts it.
48	Effect	0-1	Turns the stereo effect on or off (Chorus, Phaser or Ensemble, chosen with TYPE).

DEG 1 / DEG 2 / DEG 3 (six-stage envelopes)

No.	Name	Range	What it does
51-56	DEG 1	0-31	Volume envelope of DCO 1: Attack, Decay, Break point, Slope, Sustain, Release.
61-66	DEG 2	0-31	Volume envelope of DCO 2, same stages.
71-76	DEG 3	0-31	Filter and noise envelope, same stages.

MG (modulation generator)

No.	Name	Range	What it does
81	Frequency	0-15	MG speed (16 steps from the original ROM table).
82	Delay	0-15	Time before the MG fades in after a note starts.
83	DCO depth	0-15	Vibrato depth, up to about +/-160 cents.
84	VCF depth	0-15	Filter modulation depth.

Envelope stages (DEG 1-3)

Each DEG rises at the **Attack** rate to full level, falls at the **Decay** rate to the **Break point** level, then moves at the **Slope** rate toward the **Sustain** level, where it stays while the key is held. On release it falls at the **Release** rate. The break point and slope give the two-part decays typical of this instrument.

Host-only parameter	Range	What it does
Volume	0-1	Master output (the panel fader).
Tune	±100 cents	Fine tuning (the panel slider).
Joystick X / Joystick Y	-1 to +1	The panel joystick; automatable.
Bend Range	1-12 st	How far joystick X and the pitch wheel bend (BEND RANGE on the panel).
16 Voices	off / on	Doubles the polyphony (16 VOICES on the panel).
Key Assign	Poly / Chord	POLY or CHORD (chord memory).
Hold	off / on	Key assign HOLD (sustain pedal too).
Effect Type	Chorus / Phaser / Ensemble	Which effect P48 switches on.
Effect Speed / Intensity	0-1	Chorus or Phaser rate, or Ensemble intensity.

7. Factory programs

The 64 original factory programs, decoded from a preserved factory cassette. Numbers are bank (1-8) and program (1-8), as on the original.

11	Strings	**51**	Synthenelli
12	Acoustic Piano	**52**	Synthe Flute/Brass
13	Synthe Brass	**53**	Synthicord I
14	Flute with Chiff	**54**	Synthicord II
15	Resonant Sweep	**55**	Saxo-Brass
16	Bowed Cellos	**56**	Square Wave Ensemble
17	Full Organ	**57**	Chime Piano
18	Fat Synthe	**58**	Pipe Organ
21	Percussive Synthe	**61**	Bowed Violas
22	Muted Brass	**62**	Digi-Brass
23	Square Wave Lead	**63**	Organ Upper Manual
24	1999	**64**	Obi Brass
25	Low Strings	**65**	Soft Piano
26	Percussive Synthe (2)	**66**	Reverse Sweep
27	Bright Perc. Organ	**67**	Accordion
28	Science Bass	**68**	Steel Drums
31	Lambda	**71**	8-Note Brass
32	OB Cross	**72**	Honky Keys
33	High Strings	**73**	Organ Lower Manual
34	Double Crescendo Bass	**74**	8-Note Strings
35	Synthe Bass I	**75**	Reso Synthe
36	Reverse Bowed Synthe	**76**	Echo Synthe
37	Jazz Guitar	**77**	Black Hole
38	Helicopter	**78**	Trilling Pipes
41	Double Tongued Bass	**81**	Brass in 5ths
42	Synful Eyes	**82**	Percussive Pipe Organ
43	Wet Ones	**83**	Lunar Flutes
44	Science Strings	**84**	Synthe Bass III
45	Synthe Bass II	**85**	Vibes
46	Crescendo Brass	**86**	Low Flute
47	Chorus Piano	**87**	Who Me?
48	Jet Plane	**88**	UFO Swoop

8. About EX-800

EX-800 (called P-800 up to version 0.5.2) is built on **ISLA Poly-800** by Interspock, an open-source reconstruction of Korg's Poly-800 MkI and its rack version that recovers its modulation generator, envelope tables and factory data from the original firmware and cassette. The filter and parts of the signal path derive from Nick Copeland's Bristol synthesizer emulator.

Differences from the reconstruction it is built on

- Runs internally at 88.2/96 kHz at lower host rates, because the engine's lower-rate filter branch becomes unstable at full resonance.
- Adds joystick bend and modulation, HOLD, the hardware-style panel, and the chord memory and chorus, phaser and ensemble from Polytwelve (Ticco Ross).
- Keyboard tracking (P43) adds to the cutoff instead of replacing it, so P41 Cutoff works at every tracking setting, as on the original instrument.
- Noise (P33) is mixed at the oscillators' level; in the reconstruction it was about 19 dB too quiet to hear.
- Preset banks, with the original programs as the Factory bank.
- A new polyphonic step sequencer with parameter locks replaces the original's simple sequencer (P86-P88).
- P42 Resonance at 15 self-oscillates (a quiet, sustained tone at the cutoff while notes sound); in the reconstruction the ring always died away. 16 VOICES doubles the original polyphony.
- Not included: cassette or MIDI SysEx program transfer.

Contact

Questions, bug reports and suggestions: **ticcoross@gmail.com** (also shown under ABOUT on the panel).

Licence

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